From Liquidation Geometry to Price Movement

A Spatial-Density Heatmap and Cascade-Probability Model for Cryptocurrency Perpetual Futures

`liquidity_heatmap_preprint.tex` · 11pp · Jun 2026, revised Aug 2026 · honest methods paper — **no performance numbers anywhere**

Thesis

Leveraged perp positions carry deterministic liquidation prices. When price reaches a dense cluster of them, forced market liquidations impose impact that can reach the *next* cluster — a self-reinforcing cascade. This paper formalizes that intuition into a field, eight features, a closed-form trigger condition, and a falsifiable protocol.

The field

Discretize the log-return axis around mid into $K = 200$ bins (10 bps each, $\pm 10\%$); map each tracked position's USD notional into the bin holding its liquidation price. Eight scale- and level-invariant geometric features:

- $d_{\min}$ nearest-cluster distance · CMR mass concentration · cascade-chain length · ACP directional asymmetry
- absorption capacity A · cluster velocity · $\bar{d}$ mass-weighted distance · S spatial entropy

The mechanism

Domino condition — liquidating cluster A triggers B iff

$$\delta_B - \delta_A \; < \; \lambda \, M_A / m_t$$

(λ Kyle-type impact, M_A cluster mass, m_t depth). Order-book depth acts as a **damper** — a liquidity spiral in the Brunnermeier–Pedersen sense — and sign(ACP) predicts cascade *sign.* Geometry feeds an online, delayed-target logistic cascade-probability model.

The obstacle, as originally stated

On a decentralized venue no endpoint returns the full population of open positions. The field is a sample, $H^{\text{obs}} = \rho_t H + \text{noise}$, with ρ_t *unobserved and time-varying.* Proposed fix — shape-preserving OI rescaling:

$$H^{\text{scaled}}(t,k) = H^{\text{obs}}(t,k) \cdot \frac{\text{OI}_{\text{tot}}(t)}{\sum_k H^{\text{obs}}(t,k)}$$

leaves every ratio feature unchanged, corrects the mass-dependent ones. Original failure mode reported candidly: a preliminary multi-symbol backtest produced *zero* trades, $d_{\min}$ constantly returning its no-data sentinel.

What the Aug-2026 revision adds (§7)

ρ_t is no longer unknown — measured by the #8 instrument:

Coverage, s=2	BTC .315 / ETH .380 / SOL .247
Registry	3,596 wallets from a 41,392-wallet leaderboard
Scoping probe	equity-2k 61% OI/28% active; volume-2k 39%/46%; union 69%
Derived liq px	exact for 40.8% of 2,926 positions

Side convention s: OI counts each contract once, a registry sees both sides, so outstanding notional is 2 OI. s=2 is conservative and *is a factor of two on the one number the field is judged by.*

And the finding that changes the paper

2.9% of liquidated *wallets* vs **21.9**% of liquidated *notional*

(48 events, 35 wallets, \$223,491.) A **second bias, different in kind**: the registry is seeded by equity because the map wants whoever holds size, but whoever gets liquidated is small and levered. **The rescaling above cannot repair this** — it assumes uniform thinning, and this sampling is biased in *shape.* Cuts both ways: cascades are driven by notional and a fifth of it is not nothing; what the field cannot see is the long tail of small forced exits, which matters exactly if those *start* a chain rather than continue one.

Why the zero is not a refutation

[!] Mapped notional fraction reads 0.0%. Confounded: a liquidated wallet no longer holds the position, so a snapshot taken after the event has nothing to have predicted. Cannot yet separate "different populations" from "we looked too late". Reported as a confound, deliberately, because the bare zero reads as a refutation it has not earned.

Validation protocol status

G1–G5 are not merely unmet — they are **unevaluable** until the snapshot-then-observe series resolves. Three outcomes: mapped frac climbs (validates per-position scoring) | pinned at 0 with notional $\approx 22\%$ (demote to cluster-level scoring) | notional collapses too (seed a third registry by liquidation risk).

Falsifiable claims it makes

Cascade probability jumps discontinuously as spacing falls below $\lambda M_A / m_t$ · deep near-money depth converts approach into *rejection* rather than breakout · sign(ACP) leads cascade sign. Open question: do the eight features add anything beyond simple cluster proximity? Designed to be answerable by ablation; G5 guards against the field being a disguised liquidity proxy.

The five validation gates (pre-specified)

G1 Discrimination	OOS AUC > 0.65 *and* top-decile lift > 2×
G2 Net information	$\text{IC}_{\text{net}} = \text{IC}_{\text{gross}} - \frac{2\bar{s}}{\sigma_r \sqrt{T}} > 0.02$
G3 Stability	walk-forward median AUC > 0.60; < 20% of windows under 0.55
G4 Universality	BTC/ETH/SOL each pass G1, pairwise $\cos(\boldsymbol{\beta}_i, \boldsymbol{\beta}_j) > 0.6$
G5 Orthogonality	$R^2 < 0.30$ regressing $\hat{p}$ on a spread/depth composite

G2 makes it survive costs. **G5 is the important one**: it stops the field being a relabelled liquidity proxy — the most likely way this whole construction is secretly trivial.

What survives sampling, and what does not

Survives (shape-dependent, preserved in expectation under uniform thinning): $d_{\min}$, CMR, ACP, $\bar{d}$, S — and the largest, cascade-causing positions are disproportionately likely to be discovered since they dominate trade flow. **Biased** (absolute-mass): absorption capacity A is *over*estimated (observed mass understates the true denominator); chain length may be *under*counted when untracked positions fill gaps in the domino inequality.

Why publish results-free

Same posture as #4, #5, #6, #8: specify the mechanism, expose the observational constraint, fix the data, *then* measure. A referee can attack the construction on its merits without disentangling it from a performance claim, and there is nothing to retract when the empirics land. The §7 revision is the first return on that ordering — and it is the case for the ordering, since measuring found a bias that more modelling never would have.

Siblings

#8 is the instrument (and the source of every number in §7) · internal/confidential results-free companion `liquidity_heatmap_cascade.tex` · implementation lives at `rust/ing-features/src/heatmap.rs`.

One-pager for personal use. The lesson worth carrying: "insufficient coverage" was an *incomplete diagnosis* — coverage *of what* is the operative question, and a field can be adequate for the mass that sustains a cascade and blind to the events that start one.